Retrieval-Augmented LLMs for Culturally Sensitive Learning Content in Refugee Education



Tibakanya Joseph, Nakayiza Hellen, and Ggaliwango Marvin

Abstract Large Language Models (LLMs) have been identified as having the unrivaled potential to support educators and learners in generating high-quality educational content at various levels of education. Despite their remarkable capabilities, LLMs' concerns revolve around the lack of specific domain expertise, their propensity to generate plausible-looking content that is full of error manifested in their hallucinatory behaviors, their tendency to perpetuate bias, and their lack of knowledge temporality, which can profoundly impact the learning experiences of learners in fragile contexts such as refugee communities. This work explores leveraging Retrieval-Augmented Language Models (REALMs) in creating culturally sensitive learning content for refugee learners. Using REALMs for content generation offers an innovative and cost-effective solution that overcomes the core LLM challenges, including hallucinations, lack of domain specificity, bias, and recency issues, while addressing the unique educational challenges of marginalized communities. We discuss the applications of LLMs in creating personalized learning content, assessment item generation, automated lesson plans, schemes of work, syllabus generation, and learning content language translation. Using evidence from literature, we offer transferable use cases to refugee education. We also provide the limitations of LLMs in this task and the associated ethical concerns. Furthermore, we illuminate the processing logic of REALMs and how they overcome some limitations. Ultimately, we provide crucial insights for educators, policymakers, and AI developers on leveraging the transformative capabilities of REALMs to close the learning gaps for refugee learners and promote inclusive education, a critical aspect of Society 5.0.

Keywords Large Language Models (LLMs) · Retrieval-augmented generation (RAG) · Learning content · Refugee education

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1 Introduction

1.1 Background and Motivation

Large Language Models (LLMs), an essential advancement in Artificial Intelligence (AI), have revolutionized Natural Language Processing (NLP). The development of machine learning approaches, intense learning, the availability of high-end computational resources, and the availability of large-scale corpora all contributed to the emergence of LLMs [1]. LLMs are a type of language models (LMs) that use Neural Networks with billions of parameters that were trained on large amounts of text from websites, books, and other sources using self-supervised learning methods [2]. LLM originated in early statistical language models and is an outcome of the traditional n-gram [3]. Their lack of capacity to handle long-term interdependences and contexts in languages led to the advent of Neural Networks (NNs). Notably, the advent of Recurrent Neural Networks (RNNs) marked a significant milestone in modeling sequential data [4]. However, the challenges of vanishing gradients and long-term dependences also limited the efficacy of these RNNs. The most significant milestone occurred when the transformer architecture was introduced [1]. This introduces a self-attention mechanism, allowing parallelization and the capacity to handle long-term dependencies [5]. The architecture served as the foundation for large-scale production of pre-trained base models. For instance, OpenAI's Generative Pre-trained Transformer (GPT) series and Google's Bidirectional Encoder Representations from Transformers (BERTs) are a few examples. LLMs have become ubiquitous in most educational tasks.

They have unrivaled potential to transform education [6]. LLMs can improve learners' writing skills [7], generate assessment items [8], support intelligent tutoring, automated grading systems, and, most importantly, generate learning content [9]. Prior research has demonstrated the significant potential of LLMs in educational spaces, including the generation of lesson plans, content, question answering, formative assessment, and multiple-choice item generation.

However, using LLMs to generate such sensitive educational content is not without fault. They present an enticing avenue for developing educational content, promising unprecedented opportunities for scalability and adaptability. Despite their remarkable capabilities, LLMs' concerns revolve around the lack of specific domain expertise, their propensity to generate plausible-looking content that is full of error manifested in their hallucinatory behaviors [10, 11], their tendency to perpetuate bias [12–14], and their lack of knowledge temporality, which can profoundly impact the learning experiences of refugee learners [15].

The primary factors include the absence of locally curated, high-quality training corpora [16] and the tendency of LLM to rely on static knowledge during training until it undergoes retraining using new corpora. This process is too expensive and resource intensive. Moreover, LLMs use general-purpose corpora lacking domain-specific knowledge [14]. Consequently, the deployment of LLMs within the fragile

contexts of refugee education may face critical challenges due to their inherent limitations in comprehending complex contexts. In other words, LLMs' lack of sociocultural, linguistic, and educational backgrounds may lead to the creation of instructional materials that are inaccurate and full of misinformation, misrepresentative, or disparate, perpetuating injustices and impeding the efficacy of LLM-based educational interventions [17] in these unstable settings. To ensure the responsible deployment of LLMs, particularly in fragile educational spaces, there has been minimal effort to address and mitigate these concerns [18]. The current study seeks to answer the question: How can we leverage the transformative potential of REALMs in educational content creation while mitigating bias and hallucinatory behavior and ensuring retrieval of up-to-date content in a specific domain of refugee education?

1.2 LLMs in Educational Content Creation

The primary operational premise of LLMs involves training on diverse datasets, enabling them to acquire knowledge of intricate aspects of language, context, and domain-specific information. We enhance these models' ability to generate accurate, culturally, contextually appropriate, and logically sound instructional material by strategically prompting and fine-tuning them for tasks like answering questions. As a result, these models have exhibited a remarkable capacity to create reliable educational content in various ways.

- **Personalized Learning Content Creation:** LLMs can create educational materials specifically designed to meet each student's unique requirements, capabilities, and preferred learning methods [16]. They may enhance personalized learning by analyzing student performance data and establishing adaptive learning paths [19]. They can suggest suitable learning resources, address student inquiries, and comment on assignments [20]. This capability is crucial for learners in fragile contexts. Their learning styles are entirely different. They possess diverse linguistic capabilities, trauma levels, age, and past experiences, which affect the way they learn in the classroom.
- **Assessment Item Generation:** LLMs have shown significant potential in enhancing education by assisting teachers in generating instructional materials such as quizzes and questions for several domains, including mathematics, programming, and language learning and comprehension [21]. Several educators have leveraged the tools for multiple-choice item generation [22–27]. Most scholars noted that the produced multiple-choice questions matched the intended learning outcomes well. The researchers determine that instructors may use LLMs to capitalize on cutting-edge generative models to enhance MCQ writing endeavors [28]. Several others have utilized the tools to create open-ended questions [29–33]. This, therefore, means that LLMs can create various forms of assessment, including those requiring higher-order thinking [26]. Teachers could

leverage them to create relevant, equitable, and reliable assessments at various levels of education in diverse contexts, including refugee education.

- **Automated Content Generation:** LLMs are currently being leveraged via automation to create top-tier educational content for various use cases. For instance, automated LLM-powered systems may generate valuable information, including curriculum, syllabus, assessments, scheme of work, translated content, and lesson plans [34]. Furthermore, extensive language models may aid educators in developing comprehensive and inclusive lesson plans and activities [20]. Teachers may provide the models with the corpus of documents they wish to use as the foundation for creating a course [35]. The result might be a course syllabus with a concise summary of each subject. Language models can produce inquiries and cues that stimulate the engagement of individuals with varying knowledge and skills while prompting critical thinking and problem-solving [36]. Additionally, they can create specific and customized exercises and assessments, which may help guarantee that pupils are proficient in the subject matter.
- **Learning Content Translation:** LLMs have high-level capabilities in performing NLP tasks like language translation. Multilingual communications are common in refugee communities [37]. Teachers in such schools could harness the power of LLM to make learning content more accessible and inclusive [38]. They can translate learning materials into various languages that learners are more comfortable with. This could reduce attrition rates and minimize learning curves.
- **Curriculum Development:** LLMs may aid in curriculum design, recommending pertinent themes, learning goals, and evaluations [39]. LLMs may aid in formulating measurable learning outcomes at every stage of a curriculum and tailoring it to cater to the varied requirements of individual learners, promoting personalized and adaptable learning experiences [40]. LLMs may use performance data analysis to provide targeted interventions and advice on appropriate resources to address learning deficiencies and improve educational outcomes. In addition, incorporating LLMs into curriculum planning can assist teachers in creating, revising, or adjusting a curriculum. LLMs can recommend course material, learning goals, and pedagogical approaches based on current trends and effective methods in medical education [34]. This allows teachers to allocate more time to other teaching responsibilities. In other words, these tools can help teachers cater to the nuanced needs of learners in fragile contexts like refugee camps.

1.3 Limitations of LLMs in Educational Content Creation

Despite the remarkable potential of LLMs within the educational landscape, they suffer several limitations. The summary is provided in Fig. 1.

- **Hallucination:** LLMs exhibit a propensity to generate grammatically correct content but are full of factual errors [41–43]. This is termed a hallucination [44]. Hallucinations can be intrinsic or extrinsic. However, factuality refers to how



Fig. 1 Limitations of LLMs and ethical issues

closely LLM replies align with real-world truths and verified facts. Factuality in LLMs impacts several processes and downstream applications. This means employing such tools in fragile environments poses a significant risk.

- **Bias and Fairness:** LLMs may inadvertently generate and propagate bias that is inherent in their training data [37]. On top of that, they can generate downright harmful that can be deleterious and discriminatory [45]. This means that the generated content may not be a good fit for the diverse groups [12]. This could potentially negatively impact the learning experience of these diverse groups within the fragile educational context.
- **Lack of Domain-Specific expertise:** The massive pre-training corpora provide LLMs information but have limits. Lack of current information and specialized topic expertise are the fundamental limitations [46]. Employing LLM for educational content creation would require the LLM base models to be fine-tuned to perform that task effectively or add external information resources. Developing techniques for domain knowledge injection is critical.
- **Recency Challenges:** LLMs have a limited capacity for up-to-date knowledge besides the lack of domain-specific information. The factually correct information incorporated in LLMs has obvious temporal limitations and might become obsolete over time [46]. After training, these models do not update their internal information [45]. This is a dilemma, given our world's dynamic and ever-changing character. When faced with queries that transcend their time limit [37], LLMs often resort to falsifying facts or presenting answers that may have been valid in the past but are now obsolete [47].
- **Interpretability and Explainability:** LLMs do not explain their outputs when prompted. Understanding the decision-making process of LLMs for their outputs is a crucial issue in the field of Natural Language Processing. The capability to

interpret the LLM output is essential, especially for education in fragile environments. Firstly, it can foster trust [45]. It supports identifying and mitigating biased outputs by teachers and learners in such environments.

- **Security and Privacy:** LLM integration in numerous use cases raises worries about data privacy, **security**, and possible abuse. This is because LLMs are trained on readily available data on the Internet, some of which are private data. Yang et al. [47] advocate for LLM security and privacy awareness to increase trust.
- **Expensive Nature of Building LLMs:** LLMs are computationally costly and data-hungry, making them challenging to deploy, particularly in real-world applications with limited data and processing resources [47]. Building, training, and fine-tuning LLMs require extensive resources [4], which constitute expensive computational devices [48]. Moreover, the process has highly detrimental environmental effects due to the high energy required. This implies that retraining the models for use cases in refugee environments may not be a viable option.

2 Overview of Retrieval-Augmented Generation (RAG)

Retrieval-augmented generation (RAG) is a significant breakthrough in the field of Artificial Intelligence, namely in enhancing the capacities of LLMs [49–51]. The RAG method uses external data in real-time during the text generation process and effectively gets around many of the problems that come with traditional LLMs, such as issues with content relevance, timeliness, and factual accuracy [30]. RAG combines LLMs' fluent and coherent text generation with the capacity to collect and utilize appropriate external knowledge sources [52]. RAG can obtain and use the most relevant and up-to-date material from an internally maintained knowledge corpus, a significant benefit [51]. In short, RAG separates parametric and non-parametric information. The model parameters describe the model's interpretation and generalization of training data. External sources like vector databases, which offer updatable information, provide non-parametric knowledge. This connection allows LLMs to access domain-specific information, improving answer accuracy and relevance.

2.1 RAG Components

Retrieval Component

Retrieval refers to identifying and acquiring pertinent information in response to a specific need. Let's concentrate on information resources, which resemble a key-value store where each key corresponds to a specific value [53]. The goal is to use a similarity function to find the most similar keys to a given query and get their corresponding values [54]. The retriever plays a crucial role in the RAG system, efficiently and promptly locating pertinent documents or data from a vast collection

based on a specified query [52]. The primary function of this system is to analyze the documents and determine which ones are most relevant to the given query [49]. Retriever techniques may be classified into sparse, dense, and other categories based on various similarity functions.

Conventional retrieval techniques, like the BM25 algorithm, prioritize word frequency and presence when ranking documents, sometimes disregarding the semantic content of searches [50]. However, most current solutions use pre-trained LMs such as BERT to capture the semantic meaning of queries more efficiently [49]. These models increase the precision of search results by considering synonyms and word syntax, resulting in improved document ranking by identifying semantic resemblances [50]. Typically, this process involves calculating vector distances between pages and queries [55]. This involves combining standard retrieval measures with semantic comprehension to provide relevant search results that align with user intention.

Generation Component

The generator component is responsible for generating text relevant to the query and accurately [54] representing the information obtained from the retrieved documents [50]. The generator typically combines the query with the returned data and inputs it into an LLM for text generation [50]. While ensuring that the produced text aligns accurately with the retrieved information is challenging, it is crucial to strike a balance between strictly sticking to the original material and injecting the output with innovation [52]. The produced text must faithfully represent the information from the obtained documents and correspond to the goal of the user query [30]. Furthermore, it should facilitate the incorporation of new perspectives or insights into the gathered data [56]. We achieve these through customizing and enhancing processes [50].

2.2 RAG Performance Mechanisms

The RAG system has two fundamental components: The retriever and the generator. The retriever is responsible for searching for relevant information from the data store, usually a vector database, while the generator provides the needed content. Here's how the RAG process operates: The retriever first accepts the user input query and searches for appropriate information. Next, the generator receives the original query and retrieval results through a specified augmentation process. Lastly, the generator generates the necessary outcomes.

2.3 The Role of Prompt Engineering

Prompt engineering is a way to communicate with the LLMs. It is one of the most important skills for LLM interactions today [57]. Prompts augment LLMs with task-specific cues, allowing them to adapt to new tasks [58]. LLMs receive instructions to guarantee generating specific content qualities in the output [59]. Prompt engineering is the process of crafting, iteratively refining, and optimizing prompts to define the user's intention for the LLM [60]. Users can help LLMs bypass limitations and restrictions by meticulously designing and refining the prompts [61].

LLMs are stimulated using a variety of techniques. For instance, prompt engineering uses zero-shot or few-shot settings, eliminating the requirement for massive task-specific datasets [57]. In the few-shot scenario, the model gets one or more task samples and a natural language description, unlike in the zero-shot configuration [62]. Both approaches depend on LLMs' in-context learning capacity to accomplish a downstream job based on natural language instruction and, optionally, a few task examples [63]. In-context learning uses a prompt to condition a pre-trained model on a downstream task, whereas prompt engineering finds effective prompts. It is a form of prompting that involves exposing the model to a series of interconnected instances or prompts, allowing it to acquire knowledge and generalize based on the given context [58]. Retrieval-based prompting is another strategy that involves selecting prompts or contexts using retrieval methods [58]. This strategy involves the model retrieving pertinent prompts or context from a pool of prompts or an external knowledge base to direct its generation or decision-making process.

2.4 Benefits of REALMs in Educational Content Creation

LLMs' primary concerns are hallucinations, a lack of domain-specific knowledge, and the production of biased educational content. The leading causes revolve around the quality of training data and the failure to access up-to-date content sources. RAG tackles these challenges by retrieving relevant document chunks from an external information base via semantic similarity computations [49]. It supplements the inherent training data with updated information from outside sources, retrieving accurate information [50]. External information is conceptualized as non-parametric memory that is easy to change and contains broad, long-tail knowledge [53]. This means integrating Retrieval-Augmented LLMs (REALMs) in creating educational content for fragile educational environments such as refugee camps comes with several benefits. Firstly, it ensures the production of accurate and contextually relevant content. This is because many challenges may arise in their deployment within the fragile contexts of refugee education due to these models' inherent hallucinatory behaviors and limitations in comprehending the nuanced contexts of learners. The lack of intrinsic understanding regarding these learners' sociocultural, linguistic, and

educational backgrounds significantly hinders the efficient and ethical generation of educational content by LLMs [64].

Consequently, it may lead to the creation of instructional materials that are inaccurate, misrepresentative, or disparate. Second, the RAG technique improves the transparency of the generated outputs by allowing the model to refer to and identify data sources. This significantly promotes user confidence in automated systems [65]. Transparency plays a crucial role in establishing credibility and trustworthiness in AI-powered tools, as users often want assurance about the accuracy and origin of the information they receive. Thirdly, RAG presents more cost-effective access to a handy AI tool as an alternative to building a customized LLM due to the resources required for this task [50]. RAG uses a more dynamic approach where up-to-date educational content is integrated into the LLM via pre-trained LMs rather than fine-tuning or building an entirely new LLM [30, 66, 67]. Lastly, RAG offers LLMs a flexible and scalable feature, making them available for various tasks. For instance, if provided with similar information in its external information reservoir, they could generate lesson plans, assessment items, reading texts, and quizzes.

3 REALMs Use Cases in Educational Content Creation

Previous studies demonstrate various use cases for REALMs in educational content creation at multiple levels of education.

3.1 Challenges and Limitations of REALMs for Educational Content Creation

Despite significant advancements in RAG, there remains a need to optimize the balance between retrieval speed and the depth of accessible information [65]. This is especially important for sustaining interest in real-time interactive systems. Including RAG also raises computing performance and scalability issues, underscoring the need for more optimization.

- **The Effect of Retrieval Quality:** The introduction of RAG in LLM-generated educational content mainly ensures that the content generated is factually correct and relevant to the user's query. However, the RAG LLMs rely heavily on the retriever's quality [75]. The relevance and quality of the content stored on the retrieval source determine the output's accuracy. If the retrieval model fails to retrieve contextually relevant information relevant to the user query correctly, then the LLM produces irrelevant and possibly factually incorrect information [37]. This means that the overall dependency on the retrieval component's performance causes severe bottlenecks that require serious attention.

- **The Cost of Scalability:** RAG LLMs often require access to extensive corpora to extract pertinent information efficiently. Handling and retrieving these extensive datasets might provide considerable scaling obstacles. Deploying RAG systems, especially with extensive knowledge bases, may need significant computational resources, storage capacity, and data preparation [75]. Ensuring efficient resource use and cost-effectiveness while retaining performance is crucial for implementing RAG solutions.
- **The Complexity of Integrating with RAG-Based LLMs:** Integrating RAG-based LLM solutions with pre-existing procedures and educational technologies is challenging. The complex architectural details required to coordinate the retrieval and generation components of the RAG LLM might be overwhelming [75]. Advanced coordination mechanisms are necessary for efficient integration to ensure the smooth assimilation of acquired knowledge into the creative process. The system's intricacy may result in longer development periods, higher implementation costs, and difficulty troubleshooting and optimizing the system. Moreover, the need for multidisciplinary proficiency in retrieval techniques and generative models contributes to the complexity of integration. This can become a hurdle in environments where there is less expertise, thus hindering adoption.
- **The Effect of Domain Specificity:** RAG LLM exhibits high levels of domain specificity, indicating that its performance may drastically differ according to the individual application area. For instance, RAG LLM systems may be fine-tuned and supported for retrieving and generating medical information, but a new user may wish to use them for creating mathematical assessment items. When used in highly specialized domains, RAG systems trained on data specific to that domain may achieve high accuracy and relevance. This is because the retrieval component is more inclined to retrieve relevant information, while the generative model can generate contextually appropriate replies. Nevertheless, the narrow focus of this domain specialization might also restrict the applicability of RAG systems to broader contexts. Models trained on domain-specific datasets may require assistance in achieving high performance in broader or diverse settings because they may fail to extract or create reliable information outside their original training domain. The challenge in developing and deploying RAG LLMs is finding a balance between specialized knowledge in each topic and the capacity to apply that knowledge to other situations.
- **The Complexity of Evaluating RAG-Based LLMs:** A thorough evaluation is required when developing RAG LLMs, there is a need for thorough evaluation. Assessing the effectiveness of RAG LLMs presents distinct difficulties. The conventional measures used for retrieval, such as accuracy and recall, and generation, such as BLEU and ROUGE, may not comprehensively measure the efficacy of the RAG LLM system [75]. There is a need for assessment frameworks that comprehensively address how retrieval and generating components interact with one another. Furthermore, assessment techniques that prioritize the needs and preferences of users, like those in refugee educational environments, and measure their usefulness and satisfaction are crucial. Still, they might require significant resources and may include subjective judgements.

4 Discussion and Future Directions

This chapter's main aim is to explore how to leverage REALMs transformative potential in educational content creation while mitigating bias and hallucinatory behavior and ensuring retrieval of up-to-date content in a specific domain of refugee education. The results of this study show that using retrieval-augmented LLMs can help ensure that refugee education content is top-notch, correct, and relevant to culture and context. The analysis from Table 1 shows that REALMs technologies could be leveraged to provide personalized learning, improve assessment, provide career guidance, and improve the overall learning experiences for STEM education.

These findings emphasize the transformative potential of REALMs to revolutionize the creation of instructional materials, particularly for marginalized and heterogeneous groups of learners such as refugees. REALMs support achieving Society 5.0 aspirations by generating personalized learning content. Nevertheless, this study builds upon already established foundations by addressing the tedious task of creating appropriate learning content, especially for refugee learners. This demographic has received little attention in RAG and LLM applications. Prior research has focused on the broad applications of RAG in generating content, but this chapter explicitly highlights its significance in promoting inclusion and cultural sensitivity in educational materials.

This study has flaws despite its numerous merits, including its focus on the often-overlooked refugee community and its examination of REALMs in the educational space. The lack of empirical data and bias-free, well-curated corpora limits the research's ability to build, fine-tune, and incorporate the RAG pipeline for educational content generation. In addition, RAG-based LLM efficacy is still constrained by context window size, the cost of scaling up, a lack of expertise in these marginalized communities, and dealing with noise issues that may be the source of misinformation, among others.

Future studies should prioritize developing and verifying unbiased data collections for use in REALMs to produce culturally sensitive instructional material. As the context windows of LLMs increase enormously, it will be crucial to modify RAG frameworks to maximize their efficiency with nearly limitless context widths. It is essential to explore approaches that may improve the resilience of RAG against noise and disinformation. This includes the development of strategies to filter or reduce the influence of hostile inputs. Furthermore, research should investigate several methodologies, including RAG with fine-tuning, sequential, alternating, or end-to-end joint training, to identify the most efficient technique.

Table 1 Use cases of REALMs in educational content creation

References	Aim	Findings	Type of content	Implication
Sayed et al. [68]	Use LLMs to create a subjective question bank for k-12 education	Google Gemini LLM scored best among the models, with an average question quality score of 3.92 and a relevancy score of 0.90	Assessment items	REALMs can leverage to create subjective assessment items
Thway et al. [69]	Assess the impact of Prof. Leodar, a chatbot, on providing personalized, engaging and effective learning support while minimizing botpoop	Prof. Leodar shows a significant impact on personalized learning and student engagement in higher education	Personalized learning content, learner engagement and providing support in learning	REALMs chatbots can provide personalized learning support to learners in refugee communities
Liu et al. [70]	Assess the performance of RAG LLMs in supporting computer science education	RAG LLMs can scale learning, support content understanding and problem-solving needs	Generating programming tasks, solving problems, explaining complex topics, and fixing programming errors	REALMs can support learners taking computer science or ICT as a subject
Miladi et al. [71]	Examine the role of RAG LLMs in improving educational content accuracy	RAG-based GPT-4 increased the accuracy of learning content to about 90% from 60%	Learning content	Improves the accuracy of assessment methods and provides a more dynamic and personalized learning
Rao and Lin [72]	Provide course recommendations using RAG-based LLM for MOOCs	REALMs recommender systems can provide personalized course suggestions based on the user's needs	Personalized course recommendation information	REALMs technologies could provide career guidance to refugee learners

(continued)

Table 1 (continued)

References	Aim	Findings	Type of content	Implication
Gao et al. [73]	Exploring the potential of RAG-based LLMs in answering complex science questions	REALMs system can advance STEM education by utilizing external knowledge repositories	Answers to complex STEM questions	REALMs systems could advance STEM education in refugee education
Singh and Kumar [74]	Evaluate RAG-based LLM's efficacy in answering questions for mathematical geometry and algebra for middle school	The technologies improve the quality	Middle school question answering	REALMs technologies can improve the teaching and learning of math among refugee learners

5 Conclusion

In this work, we examined the transformative potentials of REALMs in educational content creation while mitigating bias and hallucinatory behavior and ensuring retrieval of up-to-date, culturally, and contextually relevant educational content for refugee education. We demonstrate the use of retrieval-augmented LLMs to generate equitable learning content in refugee education. We demonstrate that fine-tuning base models with high-quality bias-free corpora and supplementing the architecture with retrieval augmentation can assist in tackling bias issues, hallucinatory behaviors, and recency challenges inherent in non-task-specific LLMs. The findings from our study provide a significant guide for academics and practitioners in the field of education, particularly in challenging environments. These insights may assist them in creating REALMs for instructional material production. We anticipate that the insights from this research will drive future research and innovation in the realm of REALMs technologies, fostering equitable and inclusive learning and aiding in the achievement of SDG4. Our goal is to expedite instructional material development by incorporating retrieval techniques and language models, thereby broadening the scope of possible outcomes. We expect this work to serve as a fundamental foundation for future research endeavors, fostering the development of more sophisticated, adaptable, and contextually aware instructional material production.

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